

Markov Chains (AI HL)



Constructing a transition matrix

- 1 A weather model has two states, Sunny (S) and Rainy (R). From Sunny: 80% stay Sunny, 20% become Rainy. From Rainy: 40% become Sunny, 60% stay Rainy. Write down the transition matrix T , using columns for the 'from' state in the order (S, R).
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Applying the transition matrix once

- 2 Today is sunny, so $\mathbf{p}_0 = (1, 0)^T$ (order S, R). Using the matrix T from the previous question, find $\mathbf{p}_1 = T\mathbf{p}_0$, the probability distribution for tomorrow.
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Applying the transition matrix twice

- 3 Using $\mathbf{p}_1 = (0.8, 0.2)^T$ from the previous question, find $\mathbf{p}_2 = T\mathbf{p}_1$.
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Finding the steady-state vector

- 4 Find the steady-state vector $\mathbf{s} = (s_1, s_2)^T$ for the transition matrix T above, satisfying $T\mathbf{s} = \mathbf{s}$ and $s_1 + s_2 = 1$.
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Explaining why a Markov chain is regular

- 5 Explain why the Markov chain above converges to a unique steady state regardless of the initial distribution.
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Computing and interpreting T^2

- 6 Using the transition matrix T with rows (0.8, 0.4) and (0.2, 0.6), find T^2 and interpret its (1,1) entry.
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Constructing a 3-state transition matrix

- 7 A game has three states: Win (W), Draw (D), Loss (L). From W: 50% stay W, 30% go to D, 20% go to L. From D: 40% go to W, 20% stay D, 40% go to L. From L: 10% go to W, 30% go to D, 60% stay L. Write down the transition matrix T , using columns for the 'from' state in the order (W, D, L).
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Applying the transition matrix to a 3-state chain

- 8 Today's state is Draw, so $\mathbf{p}_0 = (0, 1, 0)^T$ (order W, D, L). Using the matrix T above, find $\mathbf{p}_1 = T\mathbf{p}_0$.
- 9 Using $\mathbf{p}_1 = (0.4, 0.2, 0.4)^T$ from the previous question, find $\mathbf{p}_2 = T\mathbf{p}_1$.
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Defining the steady-state equation

- 10 State the equation that defines a steady-state vector $\mathbf{s}$ of a transition matrix T , and explain what it means for $\mathbf{s}$ to be an eigenvector of T with eigenvalue 1.
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Verifying a steady-state vector directly

- 11 Verify that $\mathbf{s} = (2/3, 1/3)^T$ is the steady-state vector for the weather transition matrix T above, by checking $T\mathbf{s} = \mathbf{s}$ directly.
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The memoryless (Markov) property

- 12 Explain what is meant by saying a Markov chain has the 'memoryless' property.
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Constructing a new transition matrix

- 13 A machine is either Working (W) or Broken (B) each day. If Working today, there is a 90% chance it is Working tomorrow. If Broken today, there is a 30% chance it is repaired and Working tomorrow. Write down the transition matrix T , using columns for the 'from' state in the order (W, B).
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Applying the new transition matrix once

- 14 The machine is Working today, so $\mathbf{p}_0 = (1, 0)^T$ (order W, B). Using the matrix T above, find $\mathbf{p}_1 = T\mathbf{p}_0$.
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Applying the new transition matrix twice

- 15 Using $\mathbf{p}_1 = (0.9, 0.1)^T$ from the previous question, find $\mathbf{p}_2 = T\mathbf{p}_1$.
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Finding the steady state of the new chain

- 16 Find the steady-state vector $\mathbf{s} = (s_1, s_2)^T$ for the machine's transition matrix T , satisfying $T\mathbf{s} = \mathbf{s}$ and $s_1 + s_2 = 1$.
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Recognizing an invalid transition matrix

- 17 Explain why a matrix with rows $(0.9, 0.3)$ and $(0.05, 0.6)$ (order W, B) could NOT be a valid transition matrix.
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Absorbing states

- 18 A game has two states: Playing (P) and Game Over (G). From P, there is an 80% chance of remaining in P and a 20% chance of moving to G. From G, there is a 100% chance of remaining in G. Explain why G is called an 'absorbing state', and describe the long-run behaviour of the game.
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Interpreting the steady state as a long-run prediction

- 19 Using the steady-state vector $\mathbf{s} = (3/4, 1/4)^T$ found earlier for the machine example, predict the long-run fraction of days the machine is Working.
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Using a steady state to predict a count

- 20 A regular Markov chain's steady-state vector is $\mathbf{s} = (0.4, 0.35, 0.25)$ for three states A, B, C. Over 1000 independent days long after the chain has settled into steady state, roughly how many days would you expect to find the system in state B?
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Computing and interpreting T^2 for the new chain

- 21 Using the machine's transition matrix T with rows $(0.9, 0.3)$ and $(0.1, 0.7)$, find T^2 and interpret its $(2, 1)$ entry.
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Verifying the new chain's steady state directly

- 22** Verify that $\mathbf{s} = (3/4, 1/4)^T$ is the steady-state vector for the machine's transition matrix T (rows $(0.9, 0.3)$ and $(0.1, 0.7)$), by checking $T\mathbf{s} = \mathbf{s}$ directly.